TEAM ID	SWTID-2026-4021
PROJECT TITLE	India's Agricultural Crop Production Analysis
MAX MARKS	1 MARK

CHAPTER – V

PROJECT PLANNING AND SCHEDULING

5.1 – Project Milestones & Tasks

The Project Planning and Scheduling for the Agricultural Crop Production Analysis System defines the structured sequence of activities required to design, develop, test, and deploy the system. It ensures timely delivery, efficient resource utilization, and successful completion of the project within the defined scope.

Phase 1: Project Initiation & Requirement Gathering

Milestones

- Project idea approval and scope definition
- Stakeholder identification
- Requirement analysis completion

Tasks

- Study agricultural domain problems in India.
- Identify system objectives and goals.
- Gather functional and non-functional requirements.
- Analyze user needs (farmers, policymakers, researchers).
- Document system requirements.

Phase 2: System Design

Milestones

- System architecture finalized
- Database design completed

- UI/UX wireframes approved

Tasks

- Design solution architecture (layered system).
- Create Data Flow Diagrams (DFD) and use case diagrams.
- Design database schema for crop, weather, and market data.
- Develop UI mockups and dashboard layouts.
- Define technology stack.

Phase 3: Development Phase

Milestones

- Front-end development completed
- Back-end services implemented
- Database integration completed

Tasks

- Develop user interface (web/mobile).
- Implement backend logic using Python/Django or Flask.
- Build APIs for data collection and processing.
- Integrate machine learning models for crop prediction.
- Connect application with databases and external APIs.

Phase 4: Data Integration & Analytics

Milestones

- Data sources integrated
- Analytics models deployed
- Prediction system functional

Tasks

- Collect agricultural, weather, and market data.
- Clean and preprocess datasets.
- Train machine learning models for yield prediction.
- Implement data visualization dashboards.

- Validate analytics results.

Phase 5: Testing Phase

Milestones

- System testing completed
- Bug fixes finalized
- Performance validation done

Tasks

- Perform unit testing for individual modules.
- Conduct integration testing.
- Test data accuracy and prediction models.
- Perform user acceptance testing (UAT).
- Fix bugs and optimize performance.

Phase 6: Deployment Phase

Milestones

- System deployed on cloud/server
- Final version released to users

Tasks

- Deploy application on AWS/Azure/Cloud platform.
- Configure database and APIs.
- Ensure system security and backups.
- Release system to end users.

Phase 7: Maintenance & Support

Milestones

- System monitoring established
- Continuous improvement implemented

Tasks

- Monitor system performance.
- Update datasets regularly.

- Improve prediction models.
- Fix post-deployment issues.
- Provide user support and training.

Project Timeline Overview

- Requirement Gathering: 1–2 weeks
- System Design: 2 weeks
- Development: 4–6 weeks
- Data Integration & Analytics: 3–4 weeks
- Testing: 2 weeks
- Deployment: 1 week
- Maintenance: Ongoing

Conclusion

The project planning and scheduling for the Agricultural Crop Production Analysis System ensures a structured and systematic development approach. By following well-defined milestones and tasks, the project can achieve timely delivery, high-quality implementation, and effective agricultural decision support for India's farming ecosystem.